

Where does the dissipative Heisenberg chain become classically hard?

Operator entanglement and bond-dimension requirements for vectorised MPS/TEBD, and what they imply for hardware runs

Internal working note

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Abstract

We map out the region of parameter space (n, γ, t) in which a vectorised matrix-product simulation of the disordered Heisenberg chain under bulk amplitude damping stops being cheap, with the goal of choosing parameters for a quantum-hardware experiment that implements the damping via ancillas. Three results. (i) Amplitude damping produces an *entanglement barrier*: the operator entanglement rises, peaks, and decays to zero, because the dissipator’s steady state is the trivial product state $|0 \cdots 0\rangle$. The peak sits at $t^* \approx 0.45n$, *not* at $t \sim 1/\gamma$ as we initially assumed. (ii) Above a threshold system size $n_{\text{sat}}(\gamma) \approx 1.4/\gamma$ the peak becomes completely independent of n — an area law — and the classical cost stops growing, making the experiment pointless at any size. This is measured, not inferred: at $\gamma = 0.10$ the peak is 4.86, 4.89, 4.90, 4.90 bits for $n = 12, 16, 20, 24$. (iii) Along the operating line $\gamma^* = 1.4/n$, $t^* = 0.45n$, the required bond dimension already exceeds single-node feasibility at $n \approx 20$. The useful empirical law is $\chi_{\text{req}} \approx 6 \cdot 2^{1.61 S_{\text{op}}}$: the Schmidt tail costs a further factor $2^{0.6 S_{\text{op}}}$ beyond the naive $2^{S_{\text{op}}}$. We also report a methodological finding: the order-2 product formula in the existing codebase is not second order in the open case, and we give a corrected symmetric splitting. We close by stating plainly what these results do *not* establish — they bound one classical method, and the obvious competitor (quantum trajectories) is untested.

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1 Purpose and scope

The target experiment is a quantum circuit that implements the Heisenberg Hamiltonian together with an amplitude-damping channel realised through ancilla qubits. For that experiment to be worth running, the parameters (n, γ, t) must be chosen in a region where a classical simulation is *not* cheap. This note determines that region.

The scope is deliberately narrow. We study only the physical state $\rho(t)$ and the cost of representing it as a matrix-product operator. Nothing here concerns the middle-out contraction scheme, the F_{ij} Gram matrix, or multi-product formulas; those objects have a different and much less favourable cost scaling, which is why the state route reaches $n = 24$ where the F route cannot.

The reader who wants only the answer should read Sec. 8 (recommended operating points) and Sec. 9 (what this does not prove).

2 The system

2.1 Hamiltonian and dissipator

We simulate the Lindblad master equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_{j=1}^n \gamma_j \left(\sigma_j^- \rho \sigma_j^+ - \frac{1}{2} \{ \sigma_j^+ \sigma_j^-, \rho \} \right), \quad (1)$$

with the disordered anisotropic Heisenberg Hamiltonian of Eq. (13) of the main manuscript,

$$H = - \sum_{j=1}^{n-1} \left[J_j (X_j X_{j+1} + Y_j Y_{j+1}) + 2J_j Z_j Z_{j+1} \right]. \quad (2)$$

The dissipator contains *only* the lowering operator σ^- : this is amplitude damping, not a balanced emission/absorption bath. Two consequences follow immediately and shape everything below.

- **The steady state is trivial.** Eq. (1) drives the chain to $\rho_\infty = |0 \cdots 0\rangle\langle 0 \cdots 0|$, a product state with zero entanglement of any kind. Whatever entanglement the dynamics generates must therefore come back down. There is no long-time plateau to exploit, and “run longer and it gets harder” is false for this model.
- **The noise is non-unital.** Intuition imported from depolarising noise does not transfer; see Ref. [6], which shows that under non-unital noise the output distribution of a random circuit never anticoncentrates at any depth, in contrast to the unital case.

Unless stated otherwise all runs use a uniform chain $J_j = J = 0.5$ (the mean of the $\mathcal{U}[1/4, 3/4]$ disorder used elsewhere in the project) and uniform $\gamma_j = \gamma$, starting from the Néel product state. A uniform J was chosen deliberately for the scaling study: a single disorder realisation adds run-to-run noise that is easily mistaken for n -dependence.

2.2 What the gates actually implement

A caveat that matters for anyone converting our times into the units of the literature. The two-site bond gate in the codebase is $R_{ZZ}(\beta)R_{YY}(\alpha)R_{XX}(\alpha)$ with $\alpha = J_j dt$ and $\beta = 2J_j dt$, which is $\exp(-i dt h_j)$ for

$$h_j = \frac{J_j}{2}(X_j X_{j+1} + Y_j Y_{j+1}) + J_j Z_j Z_{j+1} = -\frac{1}{2}H_j^{\text{Eq. (2)}}. \quad (3)$$

The overall sign is irrelevant for entanglement, but the factor of two is not: the effective time is half what a naive reading of Eq. (2) suggests. All times quoted in this note are the code's t . The anisotropy ratio is $|J_z/J_{xy}| = 2$, which is on the mild side — Ref. [1] finds that operator entanglement *decreases* with increasing ZZ strength.

3 Simulation method

3.1 Vectorisation

We vectorise $\rho \rightarrow |\rho\rangle\rangle$ and propagate it as an MPS over n doubled sites, each carrying a ket and a bra index, so the local dimension is 4. Time evolution is TEBD: alternating odd/even Hamiltonian layers plus a single-site dissipator layer, with singular-value truncation at every two-site gate.

3.2 Bond-dimension ceilings

Three different ceilings appear in this project and confusing them is easy:

Object	Local dim.	Ceiling at bond b
$ \psi\rangle$, pure state MPS	2	$2^{\min(b, n-b)}$
$ \rho\rangle\rangle$, MPDO as MPS	4	$4^{\min(b, n-b)}$
F_{ij} , superoperator MPO	16	$16^{\min(b, n-b)}$

At $n = 20$ these are 10^3 , 10^6 and 10^{12} . The state route is what makes this study possible at all.

3.3 Trotter splitting, and a correction

The existing `get_open_step_gates` at `order=2` composes, in application order,

$$\text{odd}(dt/2), \quad \text{even}(dt), \quad \text{diss}(dt), \quad \text{odd}(dt/2) \quad \text{i.e.} \quad e^{A dt/2} e^{C dt} e^{B dt} e^{A dt/2},$$

with A, B, C the odd, even and dissipator generators. The outer layer is symmetrised but B and C are composed as a bare product, leaving an uncanceled commutator,

$$e^{C dt} e^{B dt} = \exp\left((B + C)dt + \frac{1}{2}[C, B] dt^2 + \dots\right), \quad (4)$$

so the global error carries an $O(dt)$ term whenever $[B, C] \neq 0$. For $\gamma = 0$ we have $C = 0$, the composition collapses to plain Strang and is genuinely second order — which is why closed-system testing would never have revealed this.

We measured the order directly at $n = 8$, where the MPDO ceiling $4^4 = 256$ equals the truncation cap and the run is therefore *exact*, so the only error left is the product formula. Using ratios of successive differences (which need no converged reference), Table 1 and Fig. 2(b):

The existing scheme is not cleanly first order either; it is a mixture, with the error changing *sign* between $dt = 0.05$ and $dt = 0.025$ at $t = 10$ — the fingerprint of two error terms of opposite parity. The corrected palindromic splitting

$$\text{odd}(dt/2), \text{ even}(dt/2), \text{ diss}(dt), \text{ even}(dt/2), \text{ odd}(dt/2)$$

Table 1: Effective Trotter order $p_{\text{eff}} = \log_2(\text{ratio of successive differences on halving } dt)$, measured at $n = 8$, $\gamma = 0.05$, with truncation identically zero. A clean second-order scheme gives $p_{\text{eff}} = 2$.

t	2	4	6	10
:project (existing)	1.61, 1.44	1.57, 1.39	2.30, 2.94	sign change
:strang (corrected)	2.02, 2.20	1.97, 1.96	2.00, 2.04	2.09, 2.53

is cleanly second order at every time. Being palindromic ($S(-dt) = S(dt)^{-1}$) forces the effective generator to expand in even powers of dt only.

Practically: at $t = 4$, :strang at $dt = 0.05$ is more accurate than :project at $dt = 0.025$ while taking a quarter of the steps. It costs $\sim 33\%$ more per step (one extra even layer), a net factor ~ 2.6 at fixed accuracy. All production runs reported below use :strang at $dt = 0.05$, for which the error in S_{op} relative to $dt = 0.02$ is 0.02%/0.16%/0.50% at $t = 3/6/9$.

Wider implication, not pursued here. The order-4 routine applies the Yoshida triple-jump to this same order-2 composition. That construction reaches fourth order only if the base is second order *and* palindromic, and the base is neither. Anything in the project that relies on the nominal order of `get_open_step_gates` — in particular multi-product-formula coefficient derivations, which assume an even-powers-only error expansion — should be re-examined.

3.4 The closed-system baseline

The $\gamma = 0$ control must be computed as a *pure state*, not as a density matrix. Our first attempt propagated $|\rho\rangle\rangle$ with `dissipation=false` at $\chi_{\text{max}} = 256$ and produced purities $\text{Tr}\rho^2 = 1.69, 4.19, 24.2$ at $n = 12, 16, 20$. Purity is identically 1 for a pure state, so those runs were unphysical beyond $t \approx 2.5$ and had to be discarded.

The fix uses local dimension 2 and then reconstructs the operator-space quantities *exactly*. For $\rho = |\psi\rangle\langle\psi|$ the operator Schmidt values across a cut are the pairwise products of the state’s Schmidt values $\{p_i\}$, so

$$\{\lambda^{\text{op}}\} = \{p_i p_j\}, \quad S_{\text{op}} = 2S_{\text{vN}}, \quad \chi_{\text{req}}^{\text{op}}(\epsilon) \text{ from sorting } \{p_i p_j\}. \quad (5)$$

This gives the bond dimension an MPDO code *would* have needed, from a calculation that never pays MPDO cost, and it makes purity exact by construction. Energy conservation via an MPO then provides a sharp truncation diagnostic.

4 Metrics, and why we use two

Operator entanglement S_{op} . The von Neumann entropy of the normalised squared Schmidt spectrum of $|\rho\rangle\rangle$ across a cut, in *bits*, so that $S_{\text{op}} = \log_2 \chi$ is the exact statement of “a bond dimension χ can support this much”. We report the middle cut and the maximum over cuts; measured profiles show these agree to within 4%, so the middle cut is the bottleneck.

Required bond dimension $\chi_{\text{req}}(\epsilon)$. The smallest number of Schmidt values whose discarded weight falls below ϵ (we use 10^{-6} and 10^{-10}). This, not the entropy, sets the runtime and memory.

Why both. They are not interchangeable, and the recent literature turns on the difference. Ref. [3] reports that the operator entanglement can *fall* with increasing γ while the bond dimension actually required does not, because the entropy characterises the final state whereas the simulation must carry the whole trajectory. Our data confirms the gap quantitatively: Fig. 4(c) shows

$$\chi_{\text{req}} \approx 6 \cdot 2^{1.61 S_{\text{op}}} \quad (6)$$

over three decades in χ with residual RMS 0.03 bits. The naive $2^{S_{\text{op}}}$ underestimates the cost by a factor $2^{0.61S_{\text{op}}}$, which is $\times 64$ at $S_{\text{op}} = 10$ bits. *Do not size a calculation from the entropy alone.*

Validity diagnostics. Every recorded point carries $\text{Tr}\rho$ (must be 1), $\text{Tr}\rho^2$ (must lie in $[0, 1]$ and $\rightarrow 1$ as the state relaxes to $|0 \cdots 0\rangle$), and a **saturated** flag that fires whenever χ_{max} binds at any bond. The flag matters because χ_{req} is read off the *stored* spectrum: if truncation is active, χ_{req} cannot exceed χ_{max} and the measurement is **censored**, not converged. In an early run we measured $\chi_{\text{req}} = 64, 127, 249$ at $\chi_{\text{max}} = 64, 128, 256$ — tracking the cap, not converging. Any χ_{req} from a saturated row is a lower bound only.

5 What we expected, from the literature

Closed system. Ref. [4] is the canonical statement: the MPO rank needed to represent time-evolved observables grows as $\exp(\text{const} \cdot t)$ for non-integrable chains and only polynomially when the model is integrable. Our disordered-bond chain is non-integrable; bond disorder alone does not localise (that would need random fields).

Noise as a simulability threshold. Ref. [2] established that above a characteristic system size the maximum achievable MPO entanglement is bounded by a constant depending only on the error rate, not on system size, though that bound grows rapidly as the rate decreases. Ref. [5] finds an actual transition as environment coupling increases, beyond which classical simulation succeeds at all times.

Closest prior work. Ref. [1] simulates an XXZ chain under spontaneous emission and absorption from a Néel state and compares operator entanglement (MPDO) against trajectory entanglement. It is essentially our model. Their reported behaviour: fast growth and inefficiency at $\gamma/J = 0.01$; a clear rise-and-fall at $\gamma/J \gtrsim 0.1$; and, for the maximally imbalanced case (emission only, i.e. ours), relaxation to a trivial product steady state so that $S_{\text{op}} \rightarrow 0$.

The prediction we carried in, and got wrong. From the rise-and-fall picture we assumed the barrier peaks at $t \sim 1/\gamma$ and that the hard regime was therefore *small γ at late times*. Both halves of that turned out to be wrong; see Sec. 7. The cost of the error was concrete: the first production sweep set $t_{\text{max}} \sim 2/\gamma$, which at $\gamma = 0.01$ meant integrating to $t = 200$ to capture a peak at $t = 4$, and the jobs died on walltime before reaching large n .

6 Numerical results

All runs below: uniform $J = 0.5$, Néel initial state, `:strang` splitting, $dt = 0.05$, cutoff 10^{-12} , $\chi_{\text{max}} = 256$ for the open runs.

6.1 The barrier is real, and it is transient

Fig. 1 confirms the barrier at every γ . The measured peak times are $t^* \approx 4.4\text{--}4.8$ ($\gamma = 0.10$), $4.8\text{--}8.0$ ($\gamma = 0.05$), against $1/\gamma = 10$ and 20 . The peak location is set by the *system size*, $t^* \approx 0.45n$, not by the dissipation rate, except at large γ where dissipation cuts the growth off first.

6.2 Dissipation is invisible until $\gamma t \approx 0.4$

Fig. 2(a) is the cleanest scaling result in the study: the ratio $S_{\text{op}}^{\text{open}}/S_{\text{op}}^{\text{closed}}$ collapses across $\gamma \in [0.01, 0.2]$ and remains indistinguishable from unity until $\gamma t \approx 0.4$. Before that point the amplitude damping is doing nothing measurable — the simulation is a closed-system quench.

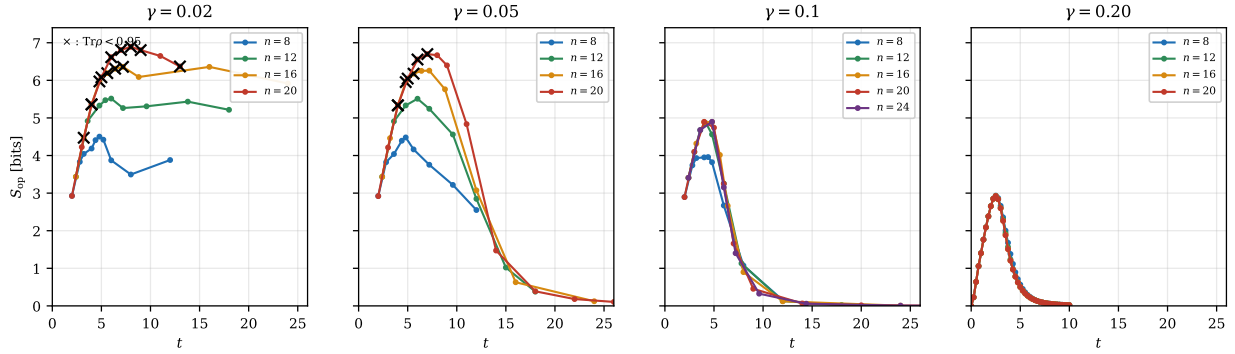


Figure 1: Operator entanglement versus time for each dissipation rate. The characteristic shape is a *barrier*: rise, peak, decay to zero, as required by the trivial product steady state of amplitude damping. Crosses mark points where $\text{Tr}\rho < 0.95$, i.e. where $\chi_{\text{max}} = 256$ was too small and the data is untrustworthy. Note the $\gamma = 0.10$ and $\gamma = 0.20$ panels, where curves for different n lie on top of one another.

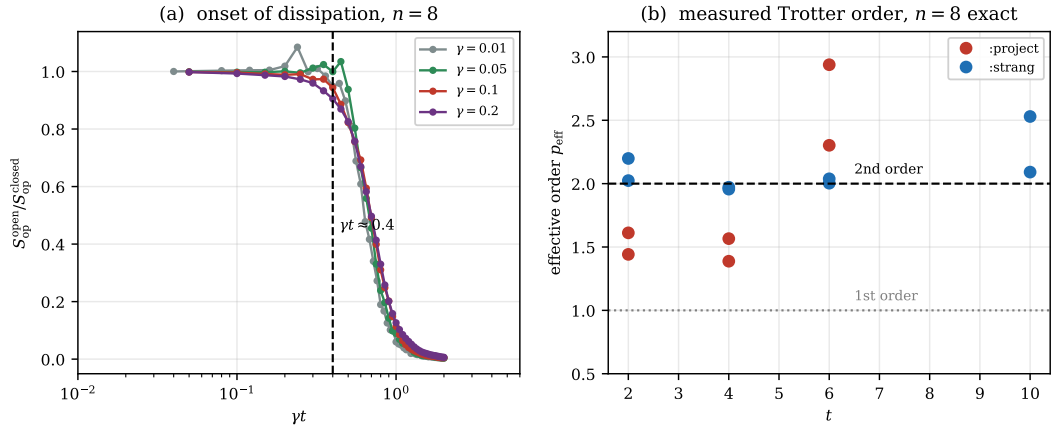


Figure 2: (a) Ratio of open to closed operator entanglement at $n = 8$, plotted against γt . Four dissipation rates spanning two decades collapse onto a single curve which stays at unity until $\gamma t \approx 0.4$ and then falls off a cliff. (b) Measured effective Trotter order (Sec. 3.3); the missing `:project` point at $t = 10$ is where its error changes sign.

Combining $t_{\text{dep}} \approx 0.4/\gamma$ with the closed-system saturation time $t_{\text{sat}} \approx n/2$ gives the condition for dissipation, rather than finite size, to cap the entanglement:

$$\frac{0.4}{\gamma} \lesssim \frac{n}{2} \iff \gamma \gtrsim \frac{0.8}{n}. \quad (7)$$

6.3 The area law, and the saturation size

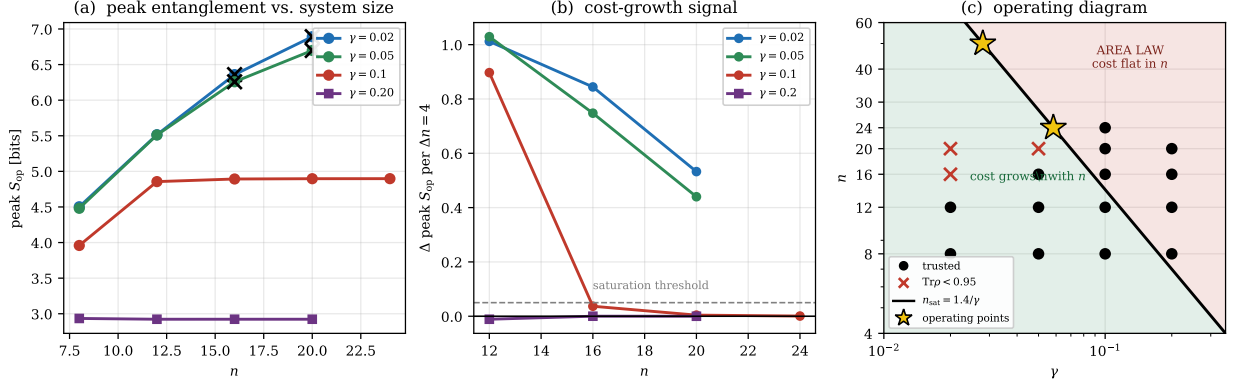


Figure 3: (a) Peak operator entanglement versus n . At $\gamma = 0.20$ and $\gamma = 0.10$ the peak stops moving; at $\gamma \leq 0.05$ it is still climbing at the largest n reached. (b) The same data as increments: zero means adding qubits is classically free. (c) Operating diagram. The line $n_{\text{sat}} = 1.4/\gamma$ is calibrated on the two saturating rates; stars mark the recommended operating points.

Table 2: Peak S_{op} [bits] versus n and γ , with the minimum trace over the run as a validity indicator. Entries with $\text{Tr}\rho < 0.95$ (bold) are truncation-corrupted and are lower bounds at best. $n = 8$ is exact (MPDO ceiling $4^4 = 256$ equals χ_{max}).

γ	$n = 8$	$n = 12$	$n = 16$	$n = 20$	$n = 24$	min $\text{Tr}\rho$ (worst n)
0.02	4.50	5.51	6.36	6.89	—	0.644
0.05	4.48	5.51	6.26	6.70	—	0.849
0.10	3.96	4.86	4.89	4.90	4.90	0.978
0.20	2.93	2.92	2.92	2.92	—	> 0.999

Table 2 contains the central result.

At $\gamma = 0.10$ the peak is 4.86, 4.89, 4.90, 4.90 bits for $n = 12, 16, 20, 24$. It does not move over a doubling of the system size. The bond-resolved profile is a flat bulk plateau with edge structure only, the trace stays above 0.978, and the purity correctly returns towards 1 as the state relaxes to $|0 \cdots 0\rangle$. This is a clean, verified area law: the classical cost is independent of n , so a hardware experiment at $\gamma = 0.10$ is classically reproducible at *any* system size.

With $\gamma = 0.20$ saturating by $n = 8$ and $\gamma = 0.10$ by $n \approx 16$, we calibrate

$$n_{\text{sat}}(\gamma) \approx \frac{1.4}{\gamma} \quad (8)$$

Below this line the peak is set by finite size; above it, by γ .

A second observation from Table 2: $\gamma = 0.02$ and $\gamma = 0.05$ are essentially *indistinguishable* at every n we reached (4.50/4.48, 5.51/5.51, 6.36/6.26). Below the saturation line the peak does not depend on γ at all, so reducing γ from 0.05 to 0.02 buys no extra classical difficulty while making the dissipation nearly invisible. There is no reason to run at $\gamma = 0.02$.

6.4 Growth laws and the cost wall

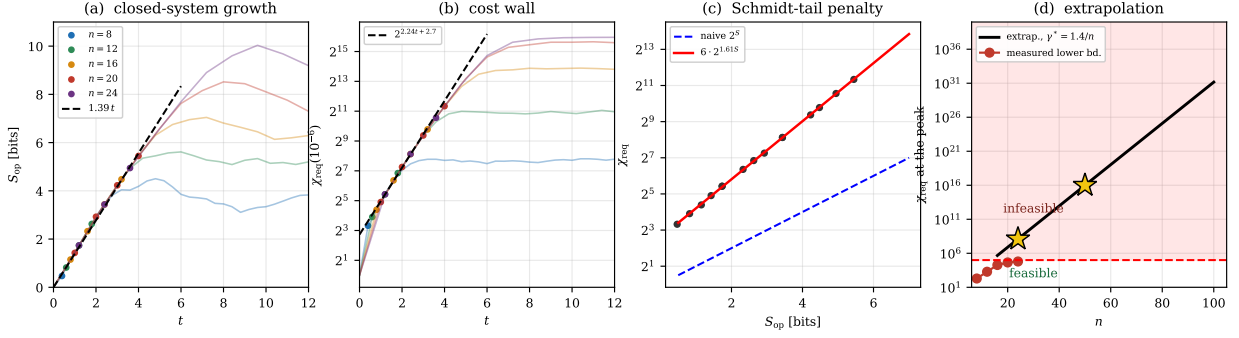


Figure 4: (a,b) Closed-system growth of S_{op} and χ_{req} . Filled markers are the pre-saturation window, where the run is exact (below both the truncation cap and the physical ceiling, and before the light cone reaches the boundary); faded lines are the truncated continuation. The curves for all n collapse, as they must while the light cone is still inside the chain. (c) The relation Eq. (6) between required bond dimension and entropy. (d) Extrapolation along the operating line, against measured lower bounds.

In the pre-saturation window the closed-system data collapses across n and gives, with per- n slopes 1.34, 1.34, 1.39, 1.51 and 2.14, 2.14, 2.24, 2.45 respectively,

$$S_{\text{op}}(t) \approx 1.39 t \text{ bits}, \quad \log_2 \chi_{\text{req}}(t) \approx 2.24 t + 2.7. \quad (9)$$

The n -independence is a consistency check rather than a result: until the light cone reaches the boundary the chain cannot know its own length. Eq. (6) follows from the ratio of the two slopes.

Table 3: **Measured** (not extrapolated) closed-system requirements at saturation. Every χ_{req} is capped by our own χ_{max} and is therefore a lower bound. Memory assumes a dense MPDO, $n \cdot 4 \cdot \chi^2 \cdot 16$ bytes.

n	S_{op} at saturation [bits]	$\chi_{\text{req}} \geq$	MPDO memory $\geq$
12	5.6	1 940	3 GB
16	7.2	18 936	367 GB
20	8.5	44 881	2.6 TB
24	10.0	62 256	6.0 TB

6.5 Extrapolation to hardware sizes

Along the operating line $\gamma^* = 1.4/n$, $t^* = 0.45n$, combining Eqs. (9) and (6):

The extrapolation is fitted on $t \lesssim 4$ and read out at $t \approx 10$ – 22 , so it should be treated as an upper envelope; growth bends over as the light cone reaches the boundary, which at $n = 50$ happens near $t \approx 25$, just past $t^* = 22.5$. Even halving the exponent, however, leaves $n = 24$ out of reach. The measured lower bounds in Table 3 make the same point without any extrapolation at all.

7 Where our initial expectations were wrong

Recorded explicitly, because two of these cost real compute time.

Table 4: Extrapolated requirement at the barrier peak along the operating line. Feasible today is roughly $\chi \sim 10^4\text{--}10^5$.

n	γ^*	t^*	S_{op} [bits]	χ_{req}	MPDO memory
16	0.088	7.2	10.0	2^{19}	10^{14} B
20	0.070	9.0	12.5	2^{23}	10^{17} B
24	0.058	10.8	15.0	2^{27}	10^{19} B
30	0.047	13.5	18.8	2^{33}	10^{23} B
50	0.028	22.5	31.3	2^{53}	10^{36} B
100	0.014	45.0	62.6	2^{104}	10^{66} B

Table 5: Corrections log.

Expected	Measured	Consequence
Barrier peaks at $t \sim 1/\gamma$; the hard regime is late times.	Peak at $t^* \approx 0.45n$, set by system size. At $\gamma = 0.01, n = 8$: peak at $t = 4, 1/\gamma = 100$.	First sweep integrated to $t = 200$ for a peak at $t = 4$; $\sim 95\%$ of walltime wasted, jobs died before large n .
Small γ is the hard regime.	γ only ever <i>reduces</i> hardness. $\gamma = 0.02$ and 0.05 give identical peaks below saturation.	“Dissipation matters” and “classically hard” are in tension; the useful region is the boundary $\gamma \approx 1.4/n$.
$\chi_{\text{max}} = 256$ ample, since S_{op} converged at $\chi = 64$.	That was measured at $\gamma = 0.2$, the easy corner. At $\gamma = 0.02, n = 20$ the trace falls to 0.644 .	$\gamma \leq 0.05, n \geq 16$ data is corrupt near the peak; only $\gamma = 0.10$ is fully trustworthy.
The order-2 splitting is second order.	$p_{\text{eff}} \approx 1.5$, with the error changing sign (Table 1).	Switched to a palindromic splitting; possible knock-on for MPF work.
Closed baseline can be run in Liouville space.	$\text{Tr}\rho^2$ reached 24.2 at $n = 20$.	Rewritten as a pure-state calculation with exact operator-space reconstruction, Eq. (5).

8 Recommended operating points

The operating line is

$$\gamma^* \approx \frac{1.4}{n}, \quad t^* \approx 0.45 n.$$

$n = 24$: $\gamma \approx 0.058$, $t \approx 11$. Already beyond single-node MPDO by the measured bounds of Table 3.

$n = 50$: $\gamma \approx 0.028$, $t \approx 22$. Extrapolated $\chi_{\text{req}} \sim 10^{16}$.

Three points deserve emphasis for whoever configures the hardware run.

1. **γ must scale with n .** Running $n > 50$ at $\gamma = 0.05$ would place the experiment *above* the saturation line ($n_{\text{sat}}(0.05) \approx 28$), in the area-law region where a classical MPDO handles any system size at fixed cost. Larger n at fixed γ is not automatically harder; past n_{sat} it is not harder at all.
2. **$t \approx 0.45n$, not the few Trotter steps used previously.** Earlier project runs used $t = 3$, which at any of these n is on the rising edge, far short of the peak.
3. **Our $dt = 0.05$ is a convergence choice, not a hardware choice.** It makes the classical run a faithful integrator of Eq. (1). The hardware will run a Trotterised circuit at whatever step the gate budget allows, which is a genuinely different map with its own entanglement growth. If the claim is to be “no classical method could match my circuit”, the classical baseline must be rerun at the hardware’s Trotter step.

9 What this does *not* establish

This study bounds one classical method. Everything above concerns the vectorised MPDO/TEBD route. The standard alternative for open systems — stochastic unravelling into quantum trajectories, i.e. pure-state MPS at local dimension 2 averaged over realisations — is completely untested here. Ref. [1] found that trajectory entanglement and operator entanglement behave differently, and can move in opposite directions as parameters vary. A regime where the MPDO route needs $\chi = 10^6$ may be one where a few thousand trajectories at $\chi = 200$ reproduce $\langle Z_j \rangle$ to within shot noise. Since the hardware experiment will be judged on observables, not on ρ , that is the comparison that decides the claim.

Secondary limitations:

- The growth laws Eq. (9) are fitted on 3–4 points per n in the window $t \lesssim 4$. The per- n consistency is reassuring but the fit deserves a dedicated large- χ_{max} closed-system run.
- $n_{\text{sat}} = 1.4/\gamma$ is calibrated on two saturating rates (0.10 and 0.20). A third, at $\gamma \approx 0.07$ with n up to 24, would test it.
- χ_{req} is not converged anywhere except $n = 8$. All larger- n values are censored lower bounds.
- All runs use a uniform J . The disorder-averaged behaviour of the original $\mathcal{U}[1/4, 3/4]$ model has not been checked.

Suggested next steps, in priority order. (1) A trajectory-entanglement study — the missing half, and cheap, since it works with pure states. (2) A closed-system-only run at large χ_{max} for $n = 24$ –30 to pin Eq. (9) out to $t \approx 12$ instead of $t \approx 4$. (3) A χ_{max} ladder at $n = 16$, $\gamma = 0.05$ to quantify how much the $\chi_{\text{max}} = 256$ truncation actually cost at the peak. (4) No further runs at $\gamma = 0.02$.

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